

Sample Surveys

Class 2

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Previously
on
Sample Surveys...

Recap

Key components

- Observation unit, Target population
- Mode of survey:
 - Complete enumeration or census, Sample survey
- Sampled population, Sampled unit, Sampling frame
- A good sample should be **representative** of the population.
- The **design** of the survey, i.e., **how the units are selected**, is very important.

Types of sampling

- Probability sampling or random sampling.
- Non-probability sampling or non-random sampling.
 - Convenience sampling
 - Judgment or purposive sampling
 - Self-selection sampling
 - Snowball sampling
 - Quota sampling

Finite population sampling framework

Running example

The state health ministry is planning to launch a new health insurance scheme. To budget for this, they must estimate the **average monthly out-of-pocket medical expenditure** for **all adults in West Bengal**.

Setup

- **Finite** population of N units: $U = \{1, 2, \dots, N\}$.
 - Sampling frame. $\rightarrow$ Units are labeled $1, 2, \dots, N$.
 - Population size: N .
- Study variable: Y (avg monthly out of pocket medical expenditure).
- Y_j : The value of the study variable for unit j in the population.
- Parameter: some feature of the population based on Y
 - Population mean: $\bar{Y} = \frac{1}{N} \sum_{j=1}^N Y_j$. (see the running example)
 - Population variance: $\sigma^2 = \frac{1}{N} \sum_{j=1}^N (Y_j - \bar{Y})^2$.
- In this setup, the Y_j s are treated as **fixed** (non-random) quantities.

Finite population sampling framework

- S : A random sample drawn from U .
- The units included in S are **random**.
- Sample size (number of units in the sample): n .
- The values of Y obtained in the sample: $y_1, y_2, \dots, y_n$.
 - For every i , y_i is Y_j for some $j \in U$.
 - $y_1, y_2, \dots, y_n$ are **random variables** $\rightarrow$ due to random sampling.
- $\mathcal{S}$: Collection of all possible samples from U . (sample space)

Sampling design

- A probability distribution over $\mathcal{S}$.
- $p: \mathcal{S} \rightarrow [0, 1]$, such that $\sum_{s \in \mathcal{S}} p(s) = 1$.
- Example: $U = \{1, 2, 3\}$. $\mathcal{S} = ?$.
 - $p(\{1, 2\}) = p(\{1, 3\}) = p(\{2, 3\}) = \frac{1}{3}$.
 - $p(s) = 0$, otherwise.

Simple Random Sampling (SRS)

- **Sampling scheme:** An algorithm/procedure to draw a sample.
- One such sampling scheme: Simple random sampling.
- The simplest form of probability sampling.
 - Select units one-by-one from the population.
 - At every stage of selection, every unit in the population has the **same marginal probability** of being selected.
- Sample size n is fixed.
- Two main types of SRS:

Simple Random Sampling *With* Replacement (SRSWR)

- 1st draw: Select one of the N units with probability $\frac{1}{N}$ each. That's the 1st unit in the sample.
- **Return** (i.e., place it back) it to the population.
- 2nd draw: Select one of the N units with probability $\frac{1}{N}$ each. That's the 2nd unit in the sample. (**the same unit can be selected again**)
- Continue like this until n units are selected.
- At every stage, the marginal probability of selecting a particular unit of the population $= \frac{1}{N}$.

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 - At every stage of selection, every unit in the population has the **same marginal probability** of being selected.
- Sample size n is fixed.
- Two main types of SRS:

Simple Random Sampling *Without* Replacement (SRSWOR)

- 1st draw: Select one of the N units with probability $\frac{1}{N}$ each. That's the 1st unit in the sample.
- **Do not return** it to the population.
- 2nd draw: Select one of the **remaining** $N - 1$ units with probability $\frac{1}{N-1}$ each. That's the 2nd unit in the sample. (the same unit **cannot** be selected again)
- Continue like this until n units are selected.
- At every stage, the **marginal** probability of selecting a particular unit of the population $= \frac{1}{N}$. (why?)

Implementation of SRS

Lottery method:



- Can be done with numbered paper folds in a box.
- Useful for small population size.
- **Troublesome** when the population size is large.

Implementation of SRS

Random number table method:

	00-04	05-09	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49
00	54463	22662	65905	70639	79365	67382	29085	69831	47058	08186
01	15389	85205	18850	39226	42249	90669	96325	23248	60933	26927
02	85941	40756	82414	02015	13858	78030	16269	65978	01385	15345
03	61149	69440	11286	88218	58925	03638	52862	62733	33451	77455
04	05219	81619	10651	67079	92511	59888	84502	72095	83463	75577
05	41417	98326	87719	92294	46614	50948	64886	20002	97365	30976
06	28357	94070	20652	35774	16249	75019	21145	05217	47286	76305
07	17783	00015	10806	83091	91530	36466	39981	62481	49177	75779
08	40950	84820	29881	85966	62800	70326	84740	62660	77379	90279
09	82995	64157	66164	41180	10089	41757	78258	96488	88629	37231

- Proposed by Fisher and Yates. Available at the back of Statistics books.
- A Million Random Digits with 100,000 Normal Deviates* (1955), RAND Corporation.

Example: SRSWR with $N = 10$, $n = 4$

- Blindly start at any row of the table. Suppose we select row 07.
- Collect the $n = 4$ single-digit numbers by moving left to right. $\rightarrow 1, 7, 7, 8$.
- Sample: Units 1,7,7,8.
- If we chose row 08, then the resulting sample: 4,10,9,5 (number 0 $\rightarrow$ unit 10).

Implementation of SRS

Random number table method:

	00-04	05-09	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49
00	54463	22662	65905	70639	79365	67382	29085	69831	47058	08186
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Example: SRSWR with $N = 17$, $n = 4$

- Blindly start at any row of the table. Suppose we select row 07.
- Collect the $n = 4$ double-digit numbers by moving left to right.
- Reject if a number is > 17 or $= 00$. Sample: Units 17,15,10,03.
- Issue: High rejection rate.

Implementation of SRS

Random number table method:

	00-04	05-09	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49
00	54463	22662	65905	70639	79365	67382	29085	69831	47058	08186
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Example: SRSWR with $N = 17$, $n = 4$

- **A fix:** Remainder method. Highest two-digit multiple of $N = 17$ is $17 \times 5 = 85$.
- Reject if a number is > 85 or $= 00$. If a number x is between 01 and 85, take $x \bmod 17$.
- If row 07 is chosen, then sample: units 17, 10, 13, 15.

Implementation of SRS

- For SRSWOR, just ignore any repeated number and proceed.
- In R, super easy to draw both SRSWR and WOR samples.

```
# Define population and sample sizes
N <- 17
n <- 4

# Draw SRSWR
srswr_draw <- sample(1:N, size = n, replace = TRUE)

# Draw SRSWOR (replace = FALSE is default)
srswor_draw <- sample(1:N, size = n, replace = FALSE)
```

- Sampling from a given data frame (say, named pop_data)

```
selected_indices <- sample(1:nrow(pop_data), size = n,
                           replace = TRUE)

sample_data <- pop_data[selected_indices, ]
```

Simple Random Sampling *With* Replacement (SRSWR)

- Typically, an SRSWR sample, one retains the **order** of selected units and their **multiplicities**.
 - E.g., (4, 3, 2, 2, 8).
- What is the **sampling design** induced by SRSWR?

SRSWR: Sampling design

- Set $\mathcal{S}$: set of all ordered samples of size n from U , allowing replacements.
- For an ordered sample $\mathbf{s} \in \mathcal{S}$,

$$p(\mathbf{s}) = \frac{1}{N^n}.$$

- The sampling design is **uniform** over $\mathcal{S}$.
- Selection probability: $\Pr(\text{Unit } j \text{ is selected in stage } k)$
 - Under SRSWR, the selection probability of each unit at each stage $= \frac{1}{N}$.
- Inclusion probability: $\Pr(\text{Unit } j \text{ is included in the sample}) =: \pi_j$.
 - Under SRSWR, $\pi_j = 1 - (1 - \frac{1}{N})^n$, for all $j \in U$.

Estimation under SRSWR

- Parameter of interest: Population mean $\bar{Y} = \frac{1}{N} \sum_{j=1}^N Y_j$.
 - Sample of Y values: $(y_1, y_2, \dots, y_n)$.
 - What is a reasonable estimator of $\bar{Y}$?
 - Sample mean: $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$.
 - Bias of the estimator: $\text{Bias}(\bar{y}) = \mathbb{E}(\bar{y}) - \bar{Y}$.
 - Turns out, $\text{Bias}(\bar{y}) = 0$, i.e., sample mean is **unbiased** for the population mean under SRSWR.
- This result holds for any other types of population means, e.g., $\frac{1}{N} \sum_{j=1}^N h(Y_j)$, with known $h(\cdot)$.
 - SRSWR sample is **representative** of the population. We can recover the population means **in expectation**.
 - **Importantly, no distributional assumptions needed** for unbiasedness. It holds **by design**.
 - All the results are similar to the case of i.i.d sampling from an infinite population.
 - Need to provide a measure of accuracy of estimation.

Estimation under SRSWR

- Population variance of Y : $\sigma^2 = \frac{1}{N} \sum_{j=1}^N (Y_j - \bar{Y})^2$.
- Turns out, $\text{Var}(\bar{y}) = \frac{\sigma^2}{n}$. $\rightarrow$ Sampling variance of $\bar{y}$.
- But σ^2 is **unknown**; it depends on $Y_1, Y_2, \dots, Y_N$.
 - Estimate it from the sample.
- Let $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$.
- $\mathbb{E}(s^2) = \sigma^2$ (why?).
- An **unbiased** estimator of the sampling variance of $\bar{y}$:

$$\widehat{\text{Var}}(\bar{y}) = \frac{s^2}{n}.$$

Simple Random Sampling *Without* Replacement (SRSWOR)

- Typically, an SRSWOR sample, one **gets rid of the order** of selected units.
 - E.g., $\{2, 3, 4, 7, 8\}$.
- What is the **sampling design** induced by SRSWOR?

SRSWOR: Sampling design

- Set $\mathcal{S}$: set of all unordered samples of size n from U , not allowing replacements.
- For an unordered sample $s \in \mathcal{S}$,

$$p(s) = \frac{1}{\binom{N}{n}}.$$

- The sampling design is **uniform** over $\mathcal{S}$.
- Selection probability: $\frac{1}{N}$ for all units, at every stage.
- Inclusion probability:
 - **First-order**: $\pi_j = P(\text{Unit } j \text{ is included in the sample}) = \frac{n}{N}$ for all $j \in U$.
 - **Second-order** $\pi_{jj'} = P(\text{Units } j \text{ and } j' \text{ are included in the sample}) = \frac{n(n-1)}{N(N-1)}$.